Presentation

https://canva.link/xizt2s9kd0tbh45

Who talks on which slides:
Katherine ->
Danni ->
Jan ->

2 min 30s intro
1 min NGO and evaluation criteria
4 min Final design
5 min diverging + testing
1 min 30s team value + position
1 min sustainability

time	slide	content	IAT
0-2 (danni)	1-4	intro - framing & opportunity	framing and stakeholder awareness
2-4 (Kath)	5-6	values & sustainability	team position/sustainability
4-7 (kath)	7	final design	
7-8 (danni)	8-12	NGO, EC	tool use (morph charts, attribute listing, pugh charts, stretches)
8 - 10 (danni)	13-14 15-20	diverging & converging if include design concepts description	representation/prototype quality
10-13 (jan)	23-25 26	verification (testing) & validation (screenshot mr.ginsberg's answer in quercus)	data accuracy/stakeholder feedback
13-14 (jan)	27	next step	
14-15(jan)	29	conclusion	holistic integration

prototypes
- the box
    - scale prototype

- the drawer
    - function prototype
- weight calculation
    - mathematical prototype
- fan
    - function protytpe

- drum feeder (danni)
    - scale prototype

- camera (jan)
    - physical
        - functional prototype
    - code
    - photo
- piano box (jan) (danni and kath do the demo
    - show demo (choose one screw before showcase)

Draft script

team number + name

slide 1 - intro

- The project is about to effect a verified and validated sustainable improvement in the lived experience of a community in the GTA.

slide 2 - the opportunity

- Woodworkers in the Greater Toronto Area frequently work with a wide variety of screws differing in both head shape and length. In small workshop environments, these screws are often stored in mixed containers, making manual sorting time-consuming, repetitive, and prone to error. This inefficiency disrupts workflow, reduces productivity, and diverts attention from higher-value tasks. This design opportunity focuses on developing an automated screw sorting system that can accurately classify screws by head shape and length.

slide 3 - the stakeholder

- Our primary stakeholder is Mr. Ginsberg, the owner of a GTA wood workshop. He works in a compact space where efficiency and organization are essential. He deals with large quantities of mixed screws regularly that need to be sorted manually.
- He needs a system that reduces manual sorting effort while maintaining high accuracy and speed. Speed is essential, as delays in organizing materials can slow down overall production. The solution must be compact to fit his limited workspace

and easy to integrate with existing tools. It should also be reliable, easy to use, and provide clear feedback during operation.

slide 4 - screw types

- Mr.Ginsberg mainly uses the Robertson screws of three different head shapes: flat head, oval head, and round washer head. Each of the three heads contains 10 different lengths, which is divided for ¾ to 3 inches with ¼ inches interval (1.9 cm to 7.6 cm).
- So the main goal is to sort these 30 types of screws plus any other rare head shape.

slide 5 - team value & position

- balabala
    - as student engineers, we initially defaulted to the “more tech is better” mindset. we assumed that high-accuracy sensors and complex computer vision for every variable were the only “professional” solution. through stakeholder Q&A and further discussion, we realized that for mr ginsberg, reliability in a dusty workshop outweighs technological complexity.
    - we want to put ourselves into mr. ginsberg’s shoe. being an insider, (we are all mr.ginsberg hurray), we understand the need for screws to be sorted neatly with high speed and accuracy. being an student engineer, we make sure the device is small, tough, easy to use, and satisfies the sustainability purpose as well as other engineering code and standard.
    - we acknowledge our
- （build for mess) value 1: we don’t build “lab toys”; we build tools that work in a real, dusty woodshop.
    - in our design - a woodshop is full of sawdust. if we used fancy electronic sensors, they would get dirty and stop working. that’s why we chose a “piano box”, which is a mechanical way to sort screws that does not care about dust.
    - how we did it - when we were using our pugh chart to pick an idea, we gave more weight to the “toughness”. We wanted something that Mr.Ginsberg would not have to fix frequently.
- (dump and go) value 2: if a tool is hard to use, people wont use it. we want our device to be able to used intuitively. (like we realized that if a machine is fast but requires 5 minutes of setup or constant monitoring, which will make it useless to a busy person like mr. ginsberg. -> allow the stakeholder to ignore the machines while it works
    - in our design - we want to make a tool that is easy to use. mr.ginsberg is busy, he doesnt want to wait around. we designed that device so that he can just dump the screws in and walk away. we also added LED lights so he can see if it’s done from meters away.
    - how we did it
        - we design the loading hatch so that the user could have place to dump the screw.
        - we actually rescoped the part that checks for rust. we realized that checking for rust takes too much time and individuals have their own

standard for it. it’s better if the woodworker just does that himself. this kept the machines fast and simple.

- (small but mighty) value 3: we don't want to waste space. We want to fit as much power as possible into a tiny box.
    - in our design - we had a strict limit of 30cm*30cm*30cm. to make it fit, we stacked everything like a tower. the screws go in at the top left, slide down the middle, and get sorted at the bottom.
- how we did it - every time we added a new part in our morph chart, we asked ourselves: “does this make the box too big? will this causes room waste” if it did, we redesigned it until it was compact
- value 4 - sustainability

slide 6 - sustainability

- value 4: we want to build a long-term used device, and with no harm to the environment
    - in our design - we prioritize low power consumption, and we planned to use disposable materials.
    - how we did it - during the converging process, we keep the requirement for sustainability in mind. We balanced the power used with other features (ex: we preferred piano box for length sorting over the camera)

slide 7 - The final design

- explain with the poster
- moving to our final design: the automated robertson screw sorting system. the process begins at the loading hatch. Once the user inputs bulk screws, the drum feeder aligns and outputs them vertically.
- the aligned screw enters the vision system, where a Raspberry Pi camera captures its head profile. our custom classification script identifies the head shape and communicates with the routing hub. if a screw is non-conforming, it is delivered to the Rejection Bin; otherwise, it proceeds to the sorting tracks.
- from here, the transition rails - which we optimized at a specific angle to reorient the screw from a vertical to a horizontal position without dropping.
- it then slides into the piano box. this is a passive sorting mechanism: as the screw rolls over, it naturally drops into the corresponding collection drawer based on its length. this design minimizes power consumption and mechanical failure points.
- finally, we integrated a ventilation fan to prevent sawdust accumulation. this is a direct response to the dusty conditions in the GTA wood workshop. the status LEDs provide immediate feedback: solid green for completion and a flashing orange alarm for error, ensuring the system is intuitive for the user even in a noisy environment.

slide 9 - Requirements

- A. Identification: The design shall automatically sort Robertson screws into distinct vessels based on length and head shape with 97.5% accuracy

- B. Efficiency and Automation: The design shall operate continuously without user interaction after putting in the screw
- C. Usability and Training: The system shall fit within a 30*30*30 cm bounding volume during all phases of operation, and The total unloaded mass of the device shall be ≤ 4.5kg

slide 10 - evaluation criteria

- We picked the five most important evaluation criteria based on the goals and Mr.Ginsberg's need for testing the design. (shown on slide)
- The design is better when …. it has higher accuracy, faster speed, more efficient use of space, higher autonomous capacity, ease of use, and less cost

slide 11 - measurement

- (shown on slide)
- The way we measure the evaluation criteria is based on the success rate, time, number of screws, total volume or weight, visibility and the money spent

slide 12 - answers from mr. Ginsberg

- Based on the original RFP, we adjusted the evaluation criteria based on Mr. Ginsberg's need. We decided to let the design sort Robertson screws only, not sorting by rustyness but length and head shape, and added cost effectiveness to help make a better fit device for him.

slide 13 - divergent ideas

- Based on the requirements of this opportunity, we divided this task into 3 different main functions: organizing screws, sorting the head shape, and sorting the length. Each individual feature could have different designs to solve each task, and the final design would be a combination of the best of all three for each function.

slide 14 - divergent tools

- We've used a few engineering tools to diverge design ideas, such as morph chart, attribute listing, root cause analysis and brainstormings. The goal was expand our solution space as much as possible before the converging process
    - Morph chart - combine different mechanical solutions for each sub-function, giving us a matrix of possible configurations
    - Attribute listing - breaking down the whole device into its attributes - material, weights, function… we identified new ways to interact with each attribute
    - Root cause analysis - anticipate why the existing design failed to meet all the requirements
    - Classical brainstorming - we can jot down all the "crazy" ideas for the design and pick the ones that fits the requirements
    - Reverse brainstorming - we asked " how could we make the more inefficient screw sorter?"
- Based on these tools, we were able to pick 3 good designs for each feature and continue working for further converging test

slide 15 - feeders

- The feeder is to take in a bunch of unorganized screws and make them into one by one aligned ordered screws
- The drum feeder: contains a rotating drum feeder to organize screws from a bulk container. As the drum rotates, screws fall into baffles and exit the feeder vertically through a conveyor belt. Prototype demo
- The vibratory bowl can organize a bulk collection of screws into one by one alignment by vibration, the screws move upward along a track, which gradually aligns them horizontally so they exit the feeder one at a time onto a conveyor belt.
- The step feeder lift screws upward in steps and pushes them onto a track where they are aligned into a single row.

slide 16 - feeder comparison

The pugh chart comparison helped us to eliminate the vibratory bowl and step feeder, based on the data from the 3 sources provided in the slide, the two evaluation criteria that related to the requirement is the speed and the volume

criteria	drum feeder	vibratory bowl	step feeder
speed	1 20 - 50 pieces per minute	-1 40 - 200 pieces per minute	0 30 - 65 pieces per minute
volume	1 Flat horizontal cylindrical with 20cm diameter × 8 cm depth - EC3	-1 Vertical cylinder of 30cm diameter × 10 cm height	-1 Cuboid of 10 × 10 × 30 cm

slide 17 - sorting head shape

- The second function needed for the device is sorting the head shape:
- The head thickness detection requires sensors above the conveyor belt to measure the screw head thickness. Different shapes produce different readings based on the height, allowing the system to sort them into three categories.
- Raspberry pi is a camera - based vision system positioned above the screw that can identify different head types such as flat, oval, or round washer heads. The camera uses machine learning for the sorting.
- Shape Fit Matching - The screw scroll down the board to see if the overall shape matches the pre-make holes. This require the length to be sorted before the head shape so the same group of screws will have the matching length

slide 18 - head shape sorting comparison

By testing and reference design

Criteria	Thickness detection	Raspberry Pi	Fit shape
volume	0 size of 10 × 5 × 5 cm	1 Size of 8.8 × 5.6 × 1.7 cm	-1 Cuboid of 10 × 10 × 30 cm
Accuracy	1 low by research and proxy testing	0 80%	-1 30 - 65 pieces per minute

slide 19 - sorting length

- The goal for this function is to sort the 10 different lengths of screws into separate drawers
- Piano box: a tilted surface with multiple holes of increasing length. As screws slide down the surface, screws fall into the hole that corresponds to its length, allowing them to be separated into different bins. show the piano box demo
- A camera system measures the screw tail length of the screw. By analyzing the captured image, the system calculates the distance from the head to the tip of the screw to determine its length category.
- Screws slide down two sticks vertically with an angle that's blocked by baffles with different lengths. Each screw falls into the appropriate slot based on its length.

slide 20 - length sorting comparison

All by testing

criteria	piano box	camera	block
speed	1 screw rolling because of gravity, less than 1 second per screw	1 need approximate 3 seconds for the camera to take picture and analyze	0 similar to piano box < 1s/screw because of gravity
accuracy	1 (but screw may wobble when sliding; threads may get stuck on the block)	1 (but with complex lighting, background shadows, or reflective surfaces on the screws, errors may occur)	-1 Not stable, slide fly off the trail
cost	1 just 3d print, no	-1 technical stuff cost	1 just wood board

	electrics needed	higher	

slide 23 - Testing - Piano box

- To verify the screw length sorting we performed different levels of proxy testing. First we created a simple prototype with cardboard. We used screws we found in Myfab and we checked that the screws fall into their respective category with high accuracy. After confirming this is a valid approach we designed and 3D printed a prototype closer to the real version and we tested it with Robertson screws retrieved from Mr. Ginsberg's workshop. In both cases we performed the tests by sorting 3 batches of 20 screws, so 60 screws in total. For each batch we measured both the time elapsed and the accuracy. We performed this testing for the following angles: 5, 10, 15, 20 degrees. The accuracy was measured as the percentage of screws falling into their respective box. We performed the verification testing using the 3d printed model. The results show that accuracy is best at 5 and 10 degrees with around 98% accuracy, the accuracy decreases the steeper the angle with 85% accuracy at 15 degrees and only 5% accuracy at 20 degrees. Therefore the obvious choice was to stay at 10 to 5 degrees, between them the 10 degree angle provides ~30% more speed, therefore the 10 degrees option was chosen.

slide 24 & 25 - Testing and verification - Raspberry Pi

- To test the idea of using a camera and machine learning for head shape sorting, we performed two different tests. For training we took 51 pictures of each screw type, so 153 pictures in total and then we used image augmentation scripts to increase data diversity. This meant that for training we had screw positions and lighting environments that were too unrealistic for real case scenarios, but that made the machine learning detection better. During the training session, after each epoch the script ran a testing script to check accuracy. Our best epoch was the last one, epoch 25, where we achieved ~70% accuracy, this is good keeping in mind that "unrealistically bad" images were used. The second test consisted of manually placing the screws in position and taking a picture one by one with the automatic analysis script running. We tested 34 screws and achieved an accuracy of ~80%. This is really good considering the amount of time and resources we had, but we are confident that we could increase that number with heavier machine learning and better lighting environment.

slide 26 - validation

- Space Constraint Met - the final volume is 28.5x28.5x29.2
- Operational Simplicity - the user only needs to pour bulk screws into the Drum Feeder.
- Performance Reliability - the system satisfies speed and accuracy priority with Mr.Ginsber's screw

- Cost Effectiveness - we tried to minimize the cost of the overall device that we prioritized lower cost full mechanical method over intelligent method ?

slide 27 - next steps

- find the best angle for the piano box
- Improve accuracy by doing heavier machine learning with Raspberry Pi
- Minimize operational noise through vibration damping
- User testing & feedback from woodworkers

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